

Name: _____

Homework: Simplification Exercises

Laws of Exponents and Radicals

Simplify the following expressions. Your final answer must not contain negative exponents. If fractional exponents appear, express your answer using radicals.

1. Simplify: $\frac{(a^{-2}b^{-3})^{-2}}{(a^3b^{-1})^{-3}}$

2. Simplify: $(x^{-2}y^5)^{-3} \cdot (x^4y^{-1})^{-2}$

3. Simplify: $\frac{(m^{-1}n^3)^{-2}}{(m^{-4}n^{-2})^3}$

4. Simplify: $\frac{(x^2y^{-3})^{-4}}{(x^{-5}y^2)^{-1}}$

5. Simplify: $\frac{(x^{-2}y^3)^{\frac{3}{2}}}{(x^4y^{-1})^{\frac{1}{2}}}$

6. Simplify: $\frac{\sqrt{x^{-4}y^{12}}}{8}$

7. Simplify: $\sqrt{\frac{x^{-8}y^4}{x^4y^{-2}}}$

8. Simplify: $\sqrt[4]{\frac{x^{-12}y^8}{\sqrt{x^4y^{-4}}}}$

9. Simplify: $\sqrt[3]{\frac{\sqrt{x^{-6}y^{12}}}{\sqrt[5]{x^{10}y^{-5}}}}$

10. Simplify: $\sqrt{\frac{\sqrt[3]{x^{-6}y^{12}}}{\sqrt[4]{x^4y^{-8}}}}$